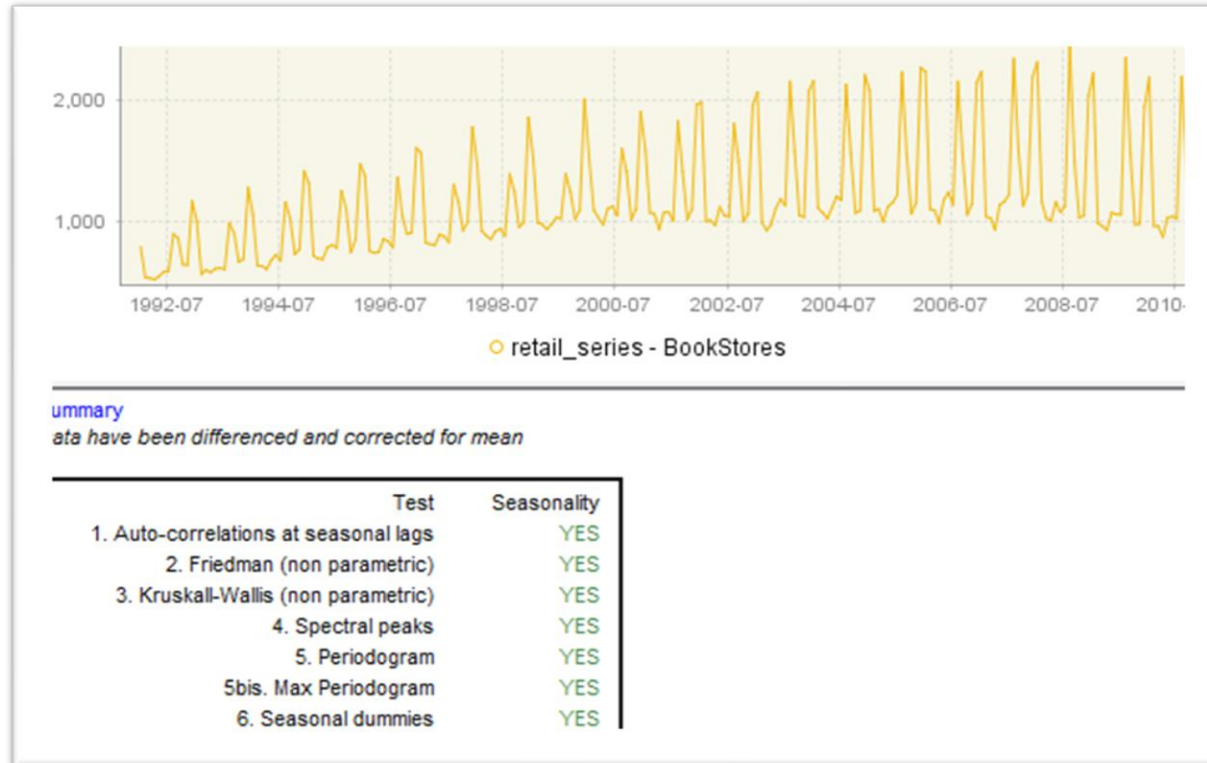


Seasonality and trading days tests in JD+

ESTP Training
05-07 October 2026

eurostat 

Overview



Seasonality present

(12)=0.9146
(24)=0.8447

Distribution: Chi2 with 2 degrees of freedom
p-value: 384.9677
p-value: 0.0000

Non parametric (Friedman) test
based on the rank of the observations in each year

Seasonality present
Distribution: Chi2 with 11 degrees of freedom
p-value: 171.0598
p-value: 0.0000

Non parametric (Kruskal-Wallis) test
based on the rank of the observations

Seasonality present
Distribution: Chi2 with 11 degrees of freedom
p-value: 188.2961
p-value: 0.0000

Identification of seasonal peaks in a Tukey periodogram and in an auto-regressive spectrum

Seasonality present

Legend: 't' for Tukey periodogram, 'A' or 'a' for auto-regressive spectrum; 'T' or 'A' for very significant peaks, 't' or 'a' for significant peaks, '_' otherwise

AT,AT,AT,AT,AT

Periodogram
based on the sum of the values of a periodogram at seasonal frequencies

Seasonality present

Distribution: F with 11 degrees of freedom in the nominator and 204 degrees of freedom in the denominator
p-value: 261.0090
p-value: 0.0000

Tests on regression with fixed seasonal dummies

Regression model (on original series) with (0 1 1)(0 0 0) noises + mean
Seasonality present

Distribution: F with 11 degrees of freedom in the nominator and 213 degrees of freedom in the denominator
p-value: 226.1038

Seasonality tests

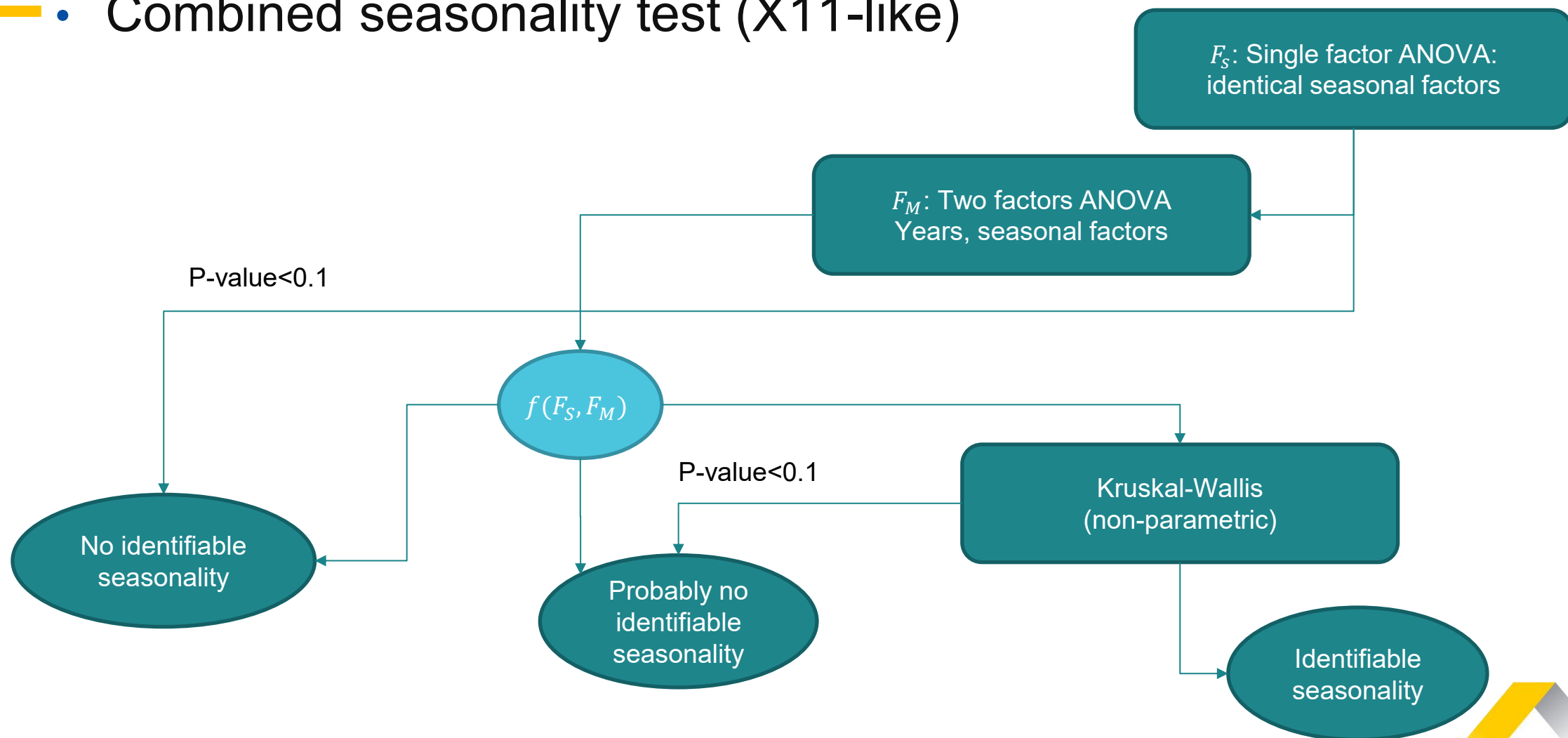
- Non-parametric tests
 - Applied on series without trend (for instance (log) differenced series)
 - **Friedman**
 - ANOVA-type. The test uses the rankings of the observations within each year. It does not require distributional assumptions.
 - H_0 : all periods can be treated equally (= no seasonality), H_1 : series is seasonal
 - **Kruskal-Wallis**
 - The test uses the rankings of all the observations. It does not require distributional assumptions.
 - H_0 : All periods have the same mean(average ranking), H_1 : series is seasonal

Seasonality tests (cont.)

- **QS test**, applied on (log) differenced series, corrected for mean effect
 - Modified Ljung-box on seasonal auto-correlations
 - H0: No correlation (no seasonality), H1: seasonality
- **F test**, applied on (log) series
 - Regression test with seasonal dummies ($D_{s,t}$) and different models
 - $y_t = \alpha + D_{s,t}\beta + \varepsilon_t$ (WN ; for residuals)
 - $y_t = \alpha + \gamma y_{t-1} + D_{s,t}\beta + \varepsilon_t$ (AR ; GUI)
 - $y_t - y_{t-1} = \alpha + D_{s,t}\beta + \varepsilon_t$ (D1; for Y, SA ; differencing $D_{s,t}$ is useless: same projection space)
 - H0: coefficients (β) are equal to 0, H1: coefficients jointly differ from 0
 - Applied on (log) series. Should be applied on parts of the series

Seasonality tests (cont.)

- Combined seasonality test (X11-like)



Seasonality tests (cont.)

- Spectral diagnostics
 - **Spectral peaks (X13-like)**
 - Based on the spectrum of a long auto-regressive model that fits the series (see X13 documentation)
 - **Periodogram**
 - Performed on the periodogram (Fourier transformation) of the series at the Fourier frequencies.
 - maximum of the periodogram on or around the seasonal frequencies
 - sum of the values of the periodogram on or around the seasonal frequencies
 - Strictly speaking, only valid against the hypothesis that the (transformed) series is a white noise. As the spectral peaks, they don't perform well for short series.

Seasonality tests (cont.)

- **Canova-Hansen test (R only)**

- test the stability of the coefficients of seasonal regressors (model quite like the F-test), using a Lagrange multiplier test.
- H_0 : stable seasonal pattern (no seasonality $\equiv$ stable 0 coefficients).
- H_1 : seasonal unit roots.
- P-values computed numerically for small samples.

Trading days tests

- **F test** on default trading days variables (contrasts)
 - Regression test with trading days contrasts (TD_t) and different models
 - $y_t = \alpha + \gamma y_{t-1} + TD_t \beta + \varepsilon_t$ (2.2.x)
 - $y_t - y_{t-1} = \alpha + TD_t \beta + \varepsilon_t$ (2.2.x)
 - $\Delta(y_t - TD_t \beta) = \alpha + \varepsilon_t$ (3.0)
 - $\Delta(y_t - TD_t \beta) = \varepsilon_t + \theta \varepsilon_{t-1}$ (3.0)
 - More generally: $y_t = \alpha + TD_t \beta + arima_t$
 - H0: coefficients (β) are equal to 0, H1: coefficients jointly differ from 0
 - Applied on (log) series. Should be applied on parts of the series

Trading days tests(cont.)

- **Canova-Hansen test** on trading days variables (**R only**)
 - test the stability of the coefficients of trading days, using a Lagrange multiplier test.
- Likelihood ratio test on time varying trading days coefficients (**R only**)
 - `rjd3toolkit` $\geq$ 3.2.5

Seasonality tests in R

```
s<-rjd3toolkit::retail$RetailSalesTotal
ls<-log(s)
st<-rjd3toolkit::do_stationary(ls, 12)
dls<-st$ddata

spec.pgram(dls)

spec.ar(dls)

print(rjd3toolkit::seasonality_qs(dls, 12))

# H0: the series has no seasonality
# pvalue = prob[x>T]
# pvalue nearly 0 -> w reject H0

print(rjd3toolkit::seasonality_kruskalwallis(dls, 12))

print(rjd3toolkit::seasonality_friedman(dls, 12))

print(rjd3toolkit::seasonality_f(ls, 12, "D1"))
```



Trading days tests in R

```
#s<-rjd3toolkit::ABS$X0.2.20.10.M
s<-rjd3toolkit::retail$RetailSalesTotal

td_all<-function(s, title, len=length(s)/12){
  a<-sapply(6:len, function(j){rjd3toolkit::td_f(s, model = "D1", nyears = j)$pvalue})
  b<-sapply(6:len, function(j){rjd3toolkit::td_f(s, model = "R011", nyears = j)$pvalue})
  c<-sapply(6:len, function(j){rjd3toolkit::td_f(s, model = "R100", nyears = j)$pvalue})

  matplot(main=title, x=6:len, y=cbind(a,b,c), pch=19, col=c('black', 'blue', 'red'),
    xlab="number of years", ylab="p-value")
}

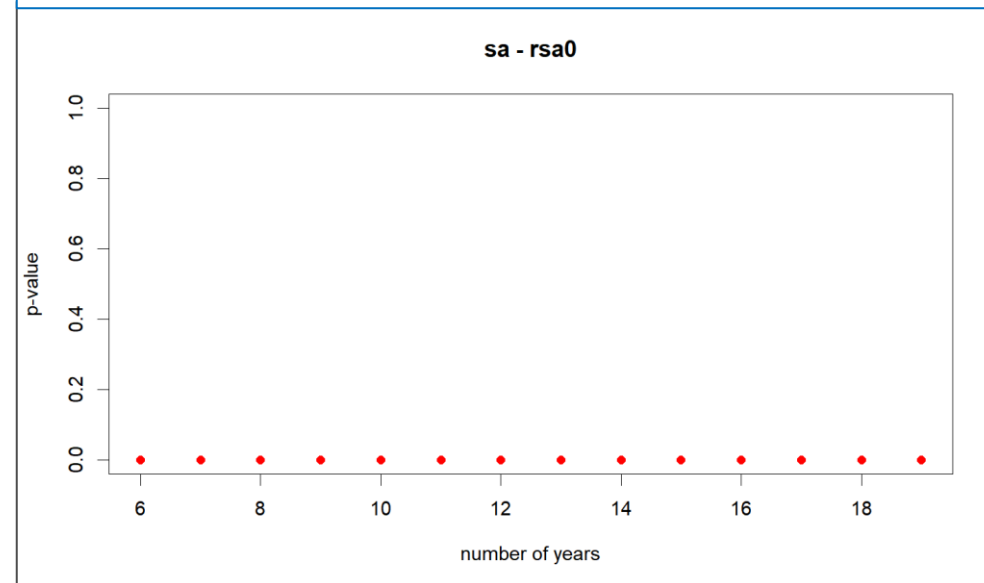
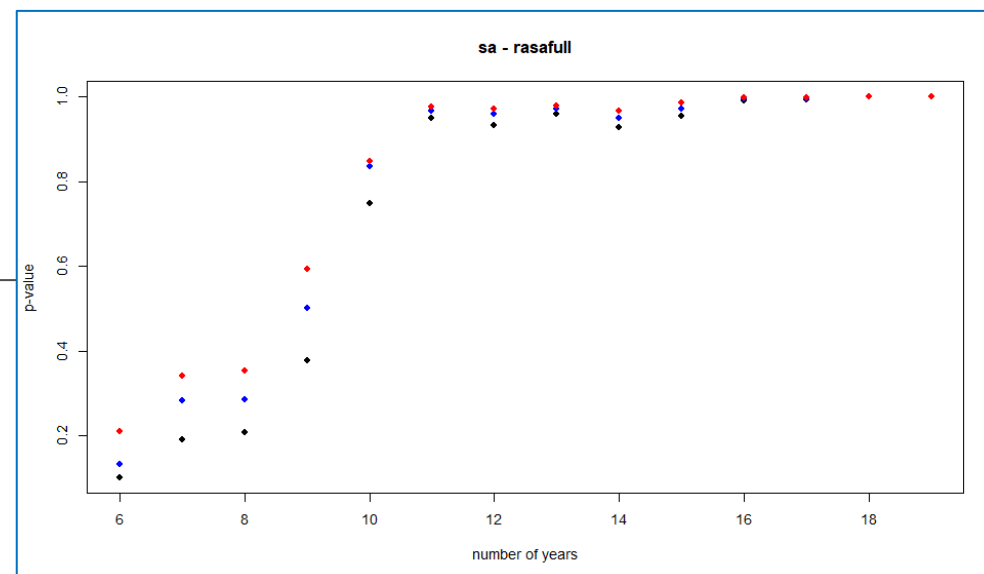
# seasonal adjustment with calendar effects correction
sa<-rjd3tramoseats::tramoseats_fast(s, "rsafull")
# seasonal adjustment without calendar effects correction
sa2<-rjd3tramoseats::tramoseats_fast(log(s), "rsa0")

ssa<-sa$decomposition$stochastics$sa$data
ssa2<-sa2$decomposition$stochastics$sa$data

td_all(ssa, "sa - rasafull")
td_all(ssa2, "sa - rsa0")

sirr<-sa$decomposition$stochastics$i$data
sirr2<-sa2$decomposition$stochastics$i$data

td_all(sirr, "irregular - rsafull")
td_all(sirr2, "irregular - rsa0")
```



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